United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Final Exam Trimester: Summer 2025 Marks: 40 Time: 2 Hours

Code: CSE 3411 Course Title: System Analysis & Design

"Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules."

Answer all the following questions:

Question 1: [CO3] [18]

Consider the following scenario:

The ATM Management System consists of two main components: the client software running on the ATM machine and a central management server located at the bank. The client software handles the user's session and transitions through various states such as *IDLE*, *CARD_INSERTED*, *PIN_ENTRY*, *AMOUNT_ENTRY*, and *TRANSACTION_IN_PROGRESS*. It interacts with different hardware modules like the **CashDispenser**, **CardReader**, and **DisplayModule**, which manage the actual physical operations of the machine. When a user initiates a withdrawal, the client software collects the account number, PIN, and desired amount, and creates a **WithdrawalRequest** object. This request is sent through a **NetworkModule**, which converts it into an ISO 8583 message and securely transmits it to the bank's central server. The server, also known as the **ATM Switch**, communicates with the **Core Banking System (CBS)** to verify the request and then returns a response—either approval or denial—back to the ATM.

Upon receiving the response, the **NetworkModule** on the client side interprets it into a clear status, such as *APPROVED*, *DENIED_INSUFFICIENT_FUNDS*, or *DENIED_INVALID_PIN*. If the transaction is approved, the ATM dispenses the specified amount of cash, displays a success message, optionally prints a receipt, and ejects the card. If the transaction is denied, the ATM displays an appropriate error message (e.g., "Insufficient Funds" or "Incorrect PIN"), ejects the card, and does not dispense any cash. After each transaction, regardless of the outcome, the system clears the session data and returns to the *IDLE* state, ready for the next user. Additionally, the bank administrator can remotely manage and maintain the ATM client, including powering it on or off.

Considering the above scenario, answer the following questions:

A) Draw the **Use Case Diagram** of the client side of the ATM, showing at least one *include* and one *extend* relationship. [4]

B) Draw the **Class Diagram** and 2 major class **CRC** card of the ATM client side. [4+2]

C) Draw the **Sequence Diagram** showing how the ATM client processes the server (ATM Switch) response to complete a transaction. [4]

D) Draw the **State Diagram** of the ATM client during a transaction. [4]

Question 2: [CO4] [5]

A) Suppose you want to launch a new mobile security software to the market. Your financial forecasting team informs you that the business will require an initial investment of $10,000, followed by additional investments after certain years. After 3 years, the company must reinvest 40% of the total revenue earned by that time. After 5 years, another investment equal to half of the previous reinvestment will be needed. In return, they expect to earn $2000, $3000, $4500 and $6000 revenue in year 2, 3, 4 and 5 respectively. The inflation rate is 15%, and the company desires a 24% return on investment.

How much revenue must the business generate in its sixth year to achieve a 24% return? Also, determine the profit or loss status of the project through the NPV Method [3]

B) Mr. Aziz wants to develop a security software. Before developing, the management has requested a feasibility analysis to determine whether the project should proceed. Briefly discuss **any two types of feasibility** relevant to this software project. [2]

Question 3: [CO3] [12]

A) SmartHome Inc. is developing a new Home Automation Mobile App. The SRS includes modules for controlling lights, temperature, and security cameras, but the energy monitoring feature requested by customers is not included. In one part of the document, it states that only homeowners can access device logs, while elsewhere it says both homeowners and guests can access logs using the same login credentials. Identify the types of SRS validation errors present in this scenario and explain the consequence of each error on system development and user experience. [2+2]

b) Automated Teller Machines (ATMs) are widely used in our daily lives for banking transactions. List some functional and nonfunctional requirements for an ATM. [3]

c) Create the proposed UI designs for the home page based on the project described in **Question 1** (client software part). Ensure the designs are user-friendly, intuitive, and visually appealing.

[3]

d) Which Golden Rules of UI Design did you consider while designing the UI for the question 3c)? [2]

Question 4: [CO3] [5]

Write down an SRS document and present it as a group having the following parts on your software project work covering: Introduction, System Study, System Analysis, Feature list fixation with Functional and Non-functional requirements, Different Feasibility Analysis, System design (UML diagrams), UI/complete protype design through Figma software, implementation plan etc. (**no need to answer this question in the examination**, it has already been evaluated in classroom activities). [5]